Blessed G/Michael Catholic Secondary School

Semester 1 Biology Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which organelle is considered the "powerhouse of the cell" responsible for ATP synthesis?

A. Golgi apparatus **B.** Ribosome

C. Mitochondrion **D.** Lysosome

____ **2.** The movement of water molecules from a region of lower solute concentration to higher solute concentration across a semi-permeable membrane is:

A. Active transport **B.** Osmosis

C. Facilitated diffusion **D.** Endocytosis

____ **3.** During which phase of mitosis do sister chromatids separate and move toward opposite poles?

A. Prophase **B.** Metaphase **C.** Anaphase **D.** Telophase

____ **4.** In which stage of the cell cycle is DNA duplicated?

A. G1 phase **B.** S phase **C.** G2 phase **D.** M phase

____ **5.** Which biological branch specifically studies heredity and variation in living organisms?

A. Ecology **B.** Genetics **C.** Cytology **D.** Physiology

____ **6.** Which structure is present in plant cells but absent in typical animal cells?

A. Cell wall and chloroplast **B.** Ribosome and nucleus

C. Plasma membrane and cytoplasm **D.** Mitochondria and vacuole

____ **7.** Which of the following microscopes uses beams of electrons rather than light to achieve magnifications up to 1,000,000x?

A. Compound light microscope **B.** Transmission electron microscope (TEM)

C. Dissecting stereomicroscope **D.** Fluorescence microscope

____ **8.** What happens to a red blood cell placed into a hypertonic salt solution?

A. It absorbs water and lyses (bursts) **B.** It loses water and crenates (shrinks)

C. It remains unchanged in shape **D.** It divides by mitosis

____ **9.** Which component of the cytoskeleton forms the mitotic spindle fibers during nuclear cell division?

A. Microfilaments (actin) **B.** Intermediate filaments

C. Microtubules (tubulin) **D.** Collagen fibers

____ **10.** What is the end product of normal mitotic cell division?

A. Four genetically distinct haploid gametes
B. Two genetically identical diploid daughter cells
C. Two haploid zygotes
D. One multinucleated clone

____ **11.** Which organelle contains hydrolytic digestive enzymes that break down worn-out cell parts and cellular debris?

A. Lysosome **B.** Ribosome **C.** Peroxisome **D.** Centrosome

____ **12.** In cell membranes, the fluid mosaic model proposes that:

A. Proteins float within a fluid phospholipid bilayer matrix
B. Proteins form a rigid solid outer shell
C. Membranes consist exclusively of nucleic acids
D. Carbohydrates form the central fluid core

____ **13.** Active transport differs from passive transport because active transport:

A. Requires metabolic energy (ATP) to move solutes against their concentration gradient
B. Does not involve membrane transport proteins
C. Occurs only in dead cells
D. Always moves water along its concentration gradient

____ **14.** What is cytokinesis in cell biology?

A. The division of cytoplasm following nuclear division (karyokinesis)
B. The condensation of chromatin into chromosomes
C. The synthesis of nuclear membrane
D. The duplication of centrioles

____ **15.** Which lens system in a compound light microscope is positioned closest to the specimen slide?

A. Objective lens **B.** Ocular (eyepiece) lens
C. Condenser lens **D.** Iris diaphragm

____ **16.** If an eyepiece has a magnification of 10x and the high-power objective is 40x, what is the total magnification?

A. 400x **B.** 50x **C.** 40x **D.** 140x

____ **17.** During which mitotic stage do chromosomes align along the equatorial plane of the cell?

A. Metaphase **B.** Anaphase **C.** Prophase **D.** Telophase

____ **18.** What is the nuclear condition of human somatic cells compared to mature gametes (sperm/egg)?

A. Somatic cells are diploid (2n = 46), gametes are haploid (n = 23)
B. Both are diploid
C. Both are haploid
D. Somatic cells are tetraploid

____ **19.** Which organelle synthesizes steroid hormones and detoxifies foreign chemical drugs in liver hepatocytes?

A. Smooth Endoplasmic Reticulum (SER) **B.** Rough Endoplasmic Reticulum (RER)
C. Lysosome **D.** Nucleolus

____ **20.** Plasmolysis in plant cells occurs when the cell is immersed in a:

A. Hypertonic solution **B.** Hypotonic solution
C. Isotonic solution **D.** Distilled water

____ **21.** Which scientist formulated the cell theory stating that all new cells originate from pre-existing cells ("Omnis cellula e cellula")?

A. Rudolf Virchow **B.** Robert Hooke
C. Anton van Leeuwenhoek **D.** Louis Pasteur

____ **22.** Centrioles are cellular structures involved in:

A. Organizing microtubule spindle fibers during animal cell division
B. Synthesizing starch granules
C. Conducting photosynthesis
D. Pumping sodium ions

____ **23.** Which cellular process engulfs large solid particles or whole bacteria by extending pseudopodia?

A. Phagocytosis ("cellular eating") **B.** Pinocytosis ("cellular drinking")
C. Exocytosis **D.** Osmosis

____ **24.** The circular region of dense RNA and protein inside the nucleus where ribosomal subunits are assembled is the:

A. Nucleolus **B.** Nuclear pore
C. Chromatin granule **D.** Nuclear lamina

____ **25.** Uncontrolled mitotic cell division resulting from mutations in cell cycle checkpoint genes leads to:

A. Cancer (tumor formation) **B.** Apoptosis
C. Tissue regeneration **D.** Cellular differentiation